

An Explainable Hybrid Deep Learning Framework for Bangla Fake News Detection Using Transformer-Based Language Models

ABSTRACT

With the rapid spread of social media and digital communication, people are now able to share and receive information very easily. However, along with this, the spread of fake news has also increased a lot, which can confuse people, create misconceptions and influence public opinion. In Bangladesh, the amount of Bengali content on Facebook, YouTube and various online news portals is increasing rapidly. Therefore, identifying fake news in Bengali has now become very important. But research on Bengali fake news detection is still relatively limited. The main reasons for this are the complexity of the Bengali language, the lack of standard datasets and the use of informal language on social media. Previously, conventional machine learning methods such as SVM and Logistic Regression were used to identify fake news. However, these methods cannot properly understand the deep meaning and context of the language.

To solve this problem, this study proposes an Explainable Hybrid Deep Learning Framework. Here, conventional machine learning techniques are used together with Transformer-based models such as BERT and XLM-RoBERTa, so that better results are obtained. In addition, Explainable AI technologies like LIME have been used to make the model's decisions easier to understand and ensure greater transparency. Experimental results of the study show that the proposed hybrid model provides higher accuracy and reliability than the single method. This research will make an important contribution to Bangla Natural Language Processing (NLP) and help in developing an effective, explainable and scalable fake news detection system for low-resource languages.

Keywords: *Deep Learning, Explainable AI, Bangla Bart, Transformer Model, Bangla Fake News.*

TABLE OF CONTENTS

CONTENTS	PAGE
Board of examiners	ii
Declaration	iii
Acknowledgements	iv
Abstract	v
CHAPTER	
CHAPTER 1: INTRODUCTION	1-6
1.1 Introduction	1-3
1.2 Motivation	3-4
1.3 Research Objectives	4
1.4 Research Questions	4-5
1.5 Expected Output	5-6
1.6 Project Management and Finance	6
1.7 Report Layout	6
CHAPTER 2: BACKGROUND	7-12
2.1 Preliminaries	7
2.2 Related Works	8-12
2.3 Research Gap	12
2.4 Challenges	13
CHAPTER 3: RESEARCH METHODOLOGY	14-25
3.1 Proposed Methodology	14-15
3.2 Dataset Details	15-17
3.3 Feature Extraction	18-20
3.3.1 Term Frequency–Inverse Document Frequency	18
3.3.2 Transformer embeddings (BERT, XLM-RoBERTa)	18
3.4 Model Analysis	18-26

3.4.1 Machine Learning Models	18
3.4.2 Hyperparameter Tuning	20
3.5 Model Development	22
3.6 Explainable AI	24
CHAPTER 4: EXPERIMENTAL RESULTS AND DISCUSSION	27-26
4.1 Evaluation Metrics	27
4.2 Performance of ML Models on Rating-Based using TF-IDF Features	29
4.3 Performance Visualization with Confusion Matrix	29
4.4 Performance Visualization with ROC Curve	30
4.5 Error Analysis — Proposed BanglaBERT Attention Model	32
4.6 Performance Visualization with LIME	33
4.7 Discussion	34
CHAPTER 5: IMPACT ON SOCIETY, ENVIRONMENT AND SUSTAINABILITY	37-43
5.1 Impact on Society	37
5.2 Impact on Environment	39
5.3 Ethical Aspects	41
5.4 Sustainability Plan	43
CHAPTER 6: CONCLUSION AND FUTURE WORK	44-48
6.1 Summary of the Study	44
6.2 Conclusions	46
6.3 Implication for Further Study	48
REFERENCES	49-51

LIST OF FIGURES

FIGURES	PAGE NO
Fig 3.1: The working process to perform Fake News Prediction	13
Fig. 3.2: Sample of the Fake News Reviews Dataset	15
Fig. 3.3: Final Class Distribution of Fake News Detection	16
Fig. 3.4: Text Length Distribution of Fake News Detection	16
Fig. 4.1: Confusion Matrix of Hybrid All Bert Model	29
Fig. 4.2: ROC Curve of Hybrid All Sentiment Labels (NR+RB+BT)	30
Fig 4.3: Error Analysis — Proposed BanglaBERT Attention Model	31
Fig. 4.4: LIME Visualization of Word Contributions for a Positive news	32
Fig. 4.5: LIME Visualization of Word Contributions for a Neutral news	32
Fig. 4.5: LIME Visualization of Word Contributions for a Negative news	33

LIST OF TABLES

TABLES	PAGE NO
Table 3.1: Best Parameters Obtained from GridSearchCV	20
Table 3.2: Best Parameters Obtained from RandomizedSearchCV	21
Table 4.2 Performance of ML Models on Rating-Based using TF-IDF Features	28

CHAPTER 1

INTRODUCTION

1.1 Introduction

In the digital age, the internet has become the main medium for exchanging information. Social media like Facebook, Twitter and various online news portals have enabled people to create, share and receive information very quickly. This has made communication and information access much easier worldwide. However, along with this, a major problem has arisen—the widespread spread of fake news. Fake news refers to false or misleading information that is presented as true news. While general misinformation is often spread unintentionally, fake news is most often deliberately created to mislead people. There may be political, financial or ideological motives behind it. The impact of fake news is very serious. It can influence people's opinions, influence elections, incite violence and reduce people's trust in the media. In the context of Bangladesh, the increase in the popularity of digital platforms has led to a rapid increase in the amount of Bengali language content. Millions of people are sharing news and opinions online every day. But due to the lack of proper regulatory mechanisms and low digital awareness, these platforms are easily becoming a means of spreading false information. Bangla fake news spreads very quickly due to emotional content, catchy headlines, and a tendency to not verify information. With the huge amount of information generated every day, it is almost impossible for humans to manually identify all fake news. Therefore, automatic fake news detection systems have become an important area of research. Such systems use Natural Language Processing (NLP) and Machine Learning technologies to determine whether the news is real or fake by analyzing text.

Initially, conventional machine learning models such as Naïve Bayes, Support Vector Machine (SVM), and Decision Tree were used to detect fake news. These models usually worked based on word usage, n-grams, and linguistic features. Although effective in some cases, these methods do not understand the deep meaning and contextual relationships of the language correctly.

Later, with the improvement of Deep Learning technology, advanced models such as Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) began to be used in fake news detection. These models can learn features directly from the text and give relatively good results. However, they also have some limitations in understanding the relationships between long sentences and the subtle meaning of language.

Currently, Transformer-based models, such as BERT (Bidirectional Encoder Representations from Transformers), have brought about a major change in the field of NLP. These models can deeply analyze the relationships between words in sentences using the self-attention method. Multilingual models like XLM-RoBERTa are able to work effectively in multiple languages, especially in Bengali.

However, Transformer-based models also have some limitations. Their computational cost is very high and it is not easy to understand how the model's decisions are made. That is, they often act as a "black box". This lack of transparency is a big problem in important areas such as fake news detection. To address these challenges, this study proposes a Hybrid Framework, where Transformer-based feature extraction is combined with a Machine Learning classifier. In addition, Explainable AI technology is used so that the model's decisions can be explained more transparently and easily. Through this, the aim is to ensure high accuracy, explainability and effectiveness.

1.2 Motivation

The main motivation for this research comes from several important challenges in the field of fake news detection, especially in the Bengali language.

First, the rapid growth of social media usage in Bangladesh has led to a significant increase in the amount of user-generated content. Platforms like Facebook are now the primary source of news for many people. However, effective mechanisms for detecting and controlling false or misleading information on these platforms are not yet robust

enough. As a result, fake news spreads very quickly and reaches a large number of people before it is verified.

Second, research on fake news detection in the Bengali language is relatively limited. Currently, most research is based on English datasets, which are not fully applicable to Bengali due to linguistic differences. The Bengali language has its own complexities, such as complex grammar, rich word formation, and the widespread use of informal language on social media. These factors make it more difficult to detect fake news in the Bengali language.

Third, conventional machine learning methods are not able to understand the deep contextual meaning of the text. Fake news often uses subtle linguistic cues, sarcastic expressions, and emotional techniques, which require advanced semantic analysis to detect accurately. Another important motivation is the need for explainability in AI systems. In real-life applications, it is not enough to simply provide accurate results; users also need to understand why the model has made a certain decision. In particular, misclassification can have serious consequences in the detection of fake news. Therefore, it is crucial to make the model's decisions transparent and easy to understand.

Finally, there is a growing demand for an effective and scalable solution that can be used in real-time. A hybrid approach combining Deep Learning and Machine Learning can balance accuracy, efficiency, and explainability. Therefore, this study proposes a hybrid framework that will make fake news detection in Bengali more reliable and effective.

1.3 Research Objectives

The main objective of this research is to design and develop an effective, reliable and interpretable framework for detecting fake news in the Bengali language. The rapid expansion of digital media and social media has made the dissemination of false and misleading information a major problem, especially in low-resource languages like Bengali. The aim of this research is to create a system using advanced computational technology that is both accurate and easy to interpret. To achieve this goal, the first step of the research will be to collect a rich Bengali fake news dataset from various reliable sources, including social media, online news portals and fact-checking websites. Then,

various preprocessing activities will be carried out to make the dataset suitable for analysis and modeling, including text cleaning, normalization, tokenization and dealing with the specific linguistic diversity of the Bengali language. Next, Transformer-based models, such as pre-trained language models, will be used to extract rich contextual features from the text. These models will be able to understand the semantic and syntactic features of the Bengali language more effectively than conventional methods.

In addition, a Hybrid Classification Framework will be created by combining Deep Learning and conventional Machine Learning methods. Through this combination, an attempt will be made to improve the overall performance of the model by using the feature representation capabilities of Deep Learning and the effective classification skills of Machine Learning together.

Standard evaluation methods such as Accuracy, Precision, Recall and F1-score will be used to verify the effectiveness and reliability of the proposed model. In addition, Explainable AI (XAI) technology will be incorporated so that users can easily understand how the model made a particular decision. This will increase user confidence in the system.

Finally, the performance of the proposed Hybrid model will be compared with conventional Machine Learning models and a single Deep Learning model, so that its effectiveness and superiority in identifying Bangla fake news can be proven in real-world situations.

1.4 Research Questions

This study was conducted based on several key research questions to analyze the effectiveness, performance, and explainability of fake news detection in Bengali. The research questions are presented simply and beautifully below:

- RQ1: How effective are Transformer-based models in fake news detection in Bengali? In particular, do they better understand the contextual, semantic, and syntactic features of the text than conventional feature extraction methods?

- RQ2: Can a hybrid classification framework that combines a deep learning model and a conventional machine learning algorithm provide better results in terms of accuracy, generalization, and computational efficiency than a single deep learning method?
- RQ3: How can explainability technology make AI-based fake news detection systems more transparent and credible? In addition, what kind of important information or insights can be obtained from the model's explanation analysis method?
- RQ4: What linguistic and contextual features play an important role in distinguishing fake news from real news in Bangla text, and how effectively can Machine Learning and Deep Learning models learn these features?
- RQ5: How does the performance of the proposed Hybrid model compare to conventional Machine Learning models and standalone Deep Learning Architectures? Specifically, how much better is its performance based on standard evaluation metrics such as Accuracy, Precision, Recall, and F1-score?

1.5 Expected Output

The expected results of this research are as follows:

- This research will develop a reliable and effective model capable of accurately identifying fake news from different sources and contexts in Bengali language texts.
- The proposed method is expected to provide better performance in terms of Accuracy, Precision, Recall and F1-score compared to conventional and baseline models.
- The research will incorporate explainability technology, so that the decision-making process of the model is more transparent and users can easily understand why a news item has been classified as fake or genuine.

- The developed system will be adaptable and scalable, so that it can be used in social media monitoring, online news verification and real-time fake news detection systems.
- This research will make an important contribution to the field of Natural Language Processing (NLP) in Bengali language. In particular, it will help provide effective methods, new ideas and reusable resources for future research.

1.6 Project Management and Finance

This research work does not receive funding from any other person or organization.

1.7 Report Layout

This thesis is organized into six chapters.

Chapter 1 presents the background of the research, motivation for the research, objectives and research questions.

Chapter 2 reviews the related studies and identifies the limitations of existing research and research gaps.

Chapter 3 describes the research methodology in detail. It discusses data collection, preprocessing, sentiment labeling, feature extraction techniques and the machine learning models used.

Chapter 4 presents the experimental results and analysis. Here, model performance, ensemble methods, hyperparameter tuning and visualization techniques such as confusion matrix, ROC curve and LIME are discussed.

Chapter 5 discusses the broader implications of the research. In particular, the importance of this research in terms of society, environment, ethical aspects and sustainable development is highlighted.

Chapter 6 presents a summary of the entire research, main results, important conclusions and possible directions for future research.

CHAPTER 2

BACKGROUND

2.1 Preliminaries

Fake news refers to false, misleading, or intentionally fabricated information that is presented as real news. Such news is usually created and disseminated with the aim of influencing public opinion, campaigning, or attracting people's attention. Detecting fake news is a difficult task, as it is often written in the same language and presentation as real news. As a result, it becomes difficult to distinguish between real and fake news without advanced analysis methods. This research mainly focuses on fake news detection in the Bengali language. There are still limitations in the Bengali language in terms of large-scale annotated datasets and advanced computational resources, which makes this research more challenging. Natural Language Processing (NLP) plays an important role in solving this problem. Through NLP, computers are able to understand, analyze, and process human language. Preprocessing techniques such as tokenization, normalization, and text cleaning are used to transform Bengali text into a structured form suitable for analysis. NLP also helps in feature extraction, which can identify important linguistic features and patterns in writing. Deep learning, an advanced branch of machine learning, makes this process more powerful. It can automatically learn complex features from large amounts of text data using multi-layer neural networks. Unlike traditional methods, it does not have to rely entirely on manual feature engineering. As a result, deep learning models are able to create high-level representations of text, which increases the accuracy and effectiveness of fake news detection systems. Transformers are a modern and powerful architecture of deep learning, which is especially important for NLP tasks. It uses self-attention mechanisms to understand the contextual relationships between words in a sentence. This allows for better analysis of the structure and meaning of sentences. Transformer-based models such as BERT have shown very successful results in various text classification tasks, including fake news detection, due to their ability to understand two-way context. Explainable AI (XAI) is an important part of modern AI systems, the

main purpose of which is to make the decision-making process of the model transparent and explainable to humans. Using XAI technology to detect fake news helps to understand why a news item is classified as fake or real. This increases user confidence and makes it easier to verify and improve the performance of the model. As a result, the system becomes more reliable and effective for real-life applications.

2.2 Related works

The rapid spread of digital media and social media has significantly increased the dissemination of fake information and fake news. As a result, automatic fake news detection is currently considered an important area of research in Natural Language Processing (NLP) and Machine Learning. Recent studies have shown that Deep Learning and Transformer-based models are the most widely used in fake news detection, as they are able to effectively understand the complex contextual relationships of text. Initially, various studies have tried to detect fake news using Deep Learning architectures such as CNN, LSTM and Bi-LSTM [1]. These models can automatically learn linguistic and semantic features, thus reducing the need for additional manual feature engineering. However, some problems such as overfitting, dataset bias and limitations in cross-domain adaptability still exist [1]. Later, researchers started working on Hybrid Deep Learning Models, where different neural network architectures are used together. For example, CNN-BiLSTM and LSTM-attention frameworks have shown good results in capturing both local linguistic features and long-term contextual dependencies [1]. In various experiments conducted using ISOT and Twitter-based fake news datasets, these hybrid models have achieved classification accuracy of over 90% [1]. However, these models require large labeled datasets and high computational resources, which pose limitations in resource-constrained environments. Recently, Transformer-based architectures have gained great importance in fake news detection. Models such as BERT, DistilBERT, and RoBERTa have been successfully used for misinformation detection in various languages and domains [2]. Comparative studies have shown that Transformer models provide

improved results in terms of accuracy and robustness compared to conventional Machine Learning and older Deep Learning models [2]. For example, Chen and Yin showed that DistilBERT achieved 99.65% accuracy and 0.996 AUC on the Kaggle Fake and True News dataset, which is much better than the CNN-LSTM model [2]. Similarly, BERT-based models have shown strong performance on COVID-19 misinformation and banking-related news datasets [2]. There have also been several studies on the efficiency and scalability of transformer models. DistilBERT, which is a compressed version of BERT, is able to provide performance close to BERT while using fewer parameters and lower computational cost [3], [4]. For example, Akpatsa et al. achieved about 94–95% accuracy using DistilBERT for COVID-19 news sentiment classification while ensuring fast training and inference times [3]. In addition, Amandeep and Suresh evaluated different DistilBERT variants and showed that the DistilBERT base-cased-distilled-squad model achieved 97.7% accuracy [4]. A comparative analysis of different Transformer architectures has also been conducted. Kitanovski et al. compared the DistilBERT and RoBERTa models and showed that while both models achieved about 99% accuracy, RoBERTa showed relatively higher generalization capability on unseen datasets [5]. Other studies have also shown good results in cross-domain performance of RoBERTa and optimized transformer models [7]. Research has been conducted using Transformer models in low-resource languages other than English. Sabado et al. studied fake news detection using BERT and DistilBERT in Filipino and achieved 94.7% accuracy using the improved DistilBERT model with data augmentation [6]. These studies show that lightweight models can be practically useful in multilingual and resource-limited environments. Ensemble learning and multi-model frameworks are also an important aspect of recent research. Ishraquzzaman et al. proposed an ensemble transformer framework combining RoBERTa and DeBERTa, which is capable of detecting fake, real, and AI-generated news [8]. Their model achieves 96.65% accuracy and uses optimization techniques such as pruning, quantization, and knowledge distillation to increase robustness [8]. Dataset quality and class imbalance also have a significant impact on fake news detection performance. Dewangga and Prasetyo analyzed the effect of dataset balancing techniques and showed that using oversampling significantly improves

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classification accuracy and recall [9]. In their study, DistilBERT trained on oversampled data achieved about 94% accuracy, which outperformed models such as CNN and Bi-LSTM [9]. Other studies have attempted to further improve fake news detection performance using various datasets, feature extraction techniques, and Deep Learning architectures [10]–[15]. These studies have highlighted the importance of combining linguistic, semantic, and contextual features. At the same time, benchmark datasets such as FakeNewsNet, LIAR, and ISOT play an important role for robust model training and cross-domain evaluation. Zhang's research shows that the ALBERT-base-v2 model achieves an F1-score of 0.908 on a dataset containing over 45,000 news articles. In addition to textual analysis, recent studies have also focused on multimodal fake news detection [17]–[22]. These methods use image, video, and social media metadata in addition to text. This allows for the detection of inconsistencies between textual and visual content, which can be useful in misinformation detection. Explainability and interpretability have also become important issues in fake news detection research [23]–[26]. Explainable Artificial Intelligence (XAI) technology helps explain which features have the most influence on the model's decision. This allows users to easily understand why a news item has been classified as fake or real. This transparency is very important in real-life applications, as automated decisions can often influence public opinion and decision-making. Although significant progress has been made in fake news detection research, some important challenges still remain. These include dataset bias, limited cross-domain generalization, the complexity of multilingual detection, and the rapid change of misinformation strategies [27]–[30]. Researchers have emphasized the need to develop more robust, scalable, and real-time misinformation detection systems. Overall, the literature review suggests that Transformer-based architectures, especially lightweight models such as DistilBERT and ALBERT, have significantly improved the performance of fake news detection. However, future research needs to focus more on improving model generalization, dataset diversity, explainability, and computational efficiency to develop more reliable and effective misinformation detection systems in the modern digital environment [30].

2.3 Research Gap

One of the major challenges in identifying fake news in Bengali is the limited availability of quality annotated datasets. The lack of sufficient and high-quality datasets makes it difficult to effectively train and evaluate powerful Machine Learning and Deep Learning models. In addition, there is a lack of effective hybrid approaches that combine Deep Learning and conventional Machine Learning methods. As a result, the opportunity to achieve better performance and good generalization by utilizing the advantages of both technologies together is limited. Another important problem is the lack of interpretability. Most advanced Deep Learning models act as “black boxes”, which makes it difficult to understand how the model reached a conclusion. This lack of transparency reduces user confidence and limits the use of the technology in real-world situations. In addition, the high computational complexity of advanced Transformer-based models acts as a major obstacle. These models require more processing power, memory, and training time, which makes it difficult to use in resource-constrained environments. Furthermore, there are limitations in establishing a direct relationship between the results of sentiment analysis and actual learning or practical development. Real-time usage, scalability and effective deployment also remain major challenges.

The main challenges of the research are summarized below:

- Limited availability of high-quality Bangla datasets
- Lack of effective Hybrid Modeling Approach
- Limitations of interpretability of Deep Learning models
- High computational complexity of the developed models
- Challenges related to deployment and scalability in real-world environments.

2.4 Challenges

Data scarcity is a major problem in fake news detection in Bengali. The lack of large, diverse, and well-annotated datasets makes it difficult to train reliable models. In addition, the inherent complexity of the Bengali language—such as rich morphology, different dialects, and contextual variations—makes text processing and feature

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extraction more difficult. Social media texts contain a lot of noise, such as slang, spelling mistakes, abbreviations, and informal language use. These elements negatively affect model performance. Another important issue is the limitation of model interpretability. Most advanced Machine Learning and Deep Learning models act as “black boxes,” making it difficult to understand how their decisions are made, and consequently, reducing trust.

In summary, the main challenges are:

- Lack of large, diverse, and balanced datasets
- Data scarcity of Bengali datasets
- Structural complexity of the Bengali language
- High levels of noise in social media texts
- Limited model interpretability

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Proposed Methodology

The proposed Bangla fake news detection framework follows a systematic process, where raw text data is gradually transformed into analytical information and ultimately accurate predictions. At the same time, emphasis is placed on ensuring the accuracy and interpretability of the system. The first step is data collection, where a comprehensive Bangla news dataset is collected from various sources. This includes online news portals, social media such as Facebook, and fact-checking websites.

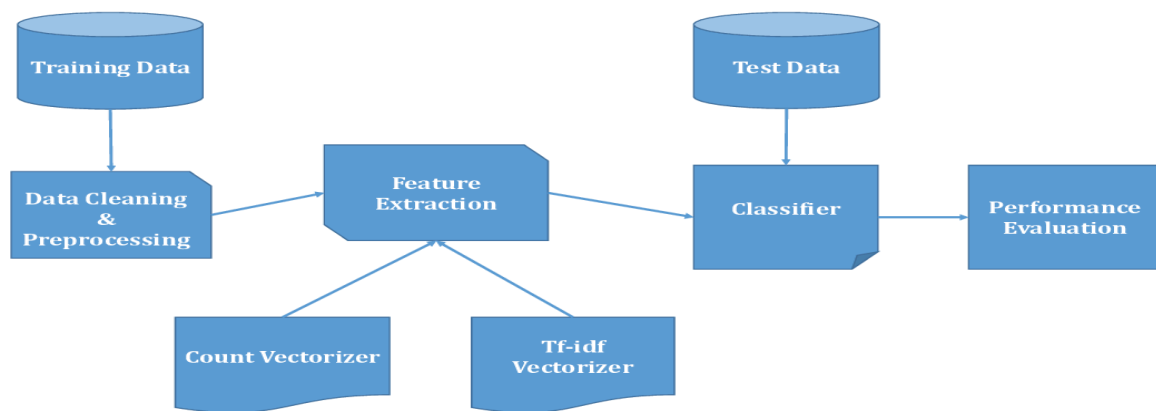


Fig 3.1: The working process to perform Fake News prediction

This dataset contains both real and fake news, so that a balanced dataset is created and effective classification is possible. Each data instance has a label along with the text, indicating whether the news is true or false. In some cases, additional information such as publication source or user engagement is also considered. The next step comes data preprocessing, which is very important for cleaning and standardizing the raw text. Online data usually contains URLs, special characters, emoji, and messy formatting. To overcome these problems, steps such as text cleaning, lowercasing, tokenization, stopwords removal and stemming/lemmatization are applied. As a result, the data becomes clearer and more suitable for analysis.

Then, in the feature extraction step, the text is converted into numbers. Various methods are used here TF-IDF, which determines the importance of words Word embeddings (Word2Vec, GloVe, FastText), which indicate the semantic relationship of words

Transformer-based models (BERT, XLM-RoBERTa), which analyze the meaning of words according to the context of the sentence Transformer-based embeddings in particular are much more effective in understanding the linguistic nuance and contextual dependency of Bengali text. Next comes the model training step, where a hybrid approach is used. Here, deep learning representations and traditional machine learning classifiers work together. The model is trained using Logistic Regression, Random Forest, XGBoost and various ensemble techniques. In addition, hyperparameter tuning is done through Grid Search and Random Search, which increases the performance and generalization ability of the model. After model training, the evaluation stage uses various standard metrics, such as accuracy, precision, recall and F1-score, to determine the overall performance of the model. In addition, the results are analyzed more clearly using visualization techniques such as confusion matrix and ROC curve.

Finally, to ensure interpretability, Explainable AI (XAI) has been added, specifically using LIME (Local Interpretable Model-Agnostic Explanations). It shows which words or features have the most impact on each prediction, so users can easily understand why a news item has been classified as fake or real. Overall, this integrated approach helps to create a robust, accurate and explainable Bangla fake news detection system, which is usable and effective in real life.

3.2 Dataset Details

The dataset used in this study is the BanFakeNews dataset, which is collected from Kaggle and was originally presented as a benchmark dataset for Bangla fake news detection at the LREC 2020 conference. It is a dataset specifically designed for low-resource language research, which includes a large number of annotated Bangla news articles. This dataset consists of multiple CSV files, namely Authentic-48K.csv, Fake-1K.csv, LabeledAuthentic-7K.csv and LabeledFake-1K.csv. The major part of it is

provided as unlabeled data and the minor part is provided as labeled data. The main dataset used in this study contains about 50,000 news articles, of which about 48,000 are authentic (true news) and about 1,000 are fake news (fake news). This indicates a clear class imbalance, which needs to be handled carefully during model training. Each data instance contains several important attributes, such as article ID, news domain, publication date, category, headline, content, and a label indicating whether the news is real (1) or fake (0).

The labeled subset also includes additional information, such as source verification, headline-content relation, and fake news type, which provides a deeper understanding of the nature of misinformation. This dataset covers a variety of news categories, such as national, international, sports, entertainment, finance, and politics. As a result, it contains diverse content and linguistic patterns. The news is collected from various popular Bengali news portals such as jagonews24.com, kalerkantho.com, and other sources, reflecting the real-world news environment.

articleID	domain	date	category	source	relation	headline	content	label	
0	1	bd-pratidin.com	2018-09-20 08:16:43	Sports	আফগান ক্রিকেট বোর্ড (এসিবি) প্রধান	Related	হঠাৎ আফগান ক্রিকেট বোর্ড প্রধানের পদত্যাগ	ক্রিকেট বিশ্বের নতুন চমকের নাম আফগানিস্তান। কর..	1
1	2	jugantor.com	2018-09-20 20:20:20	Sports	Reporter	Related	টস হেরে বোলিংয়ে বাংলাদেশ	এশিয়া কাপের ষষ্ঠ ম্যাচে বাংলাদেশ দলের বিপক্ষে ...	1
2	3	bd24live.com	2018-09-20 16:39:40	National	Reporter	Related	রাজধানীতে বিশেষ অভিযানে আটক ৪৩	রাজধানীতে মাদক বিক্রেতী বিশেষ অভিযান পরিচালনা কর..	1

Fig. 3.2: Sample of the Fake News Reviews Dataset

Temporal analysis of the dataset shows that it spans different time periods, capturing different writing styles and changes in news presentation over time. In addition, exploratory data analysis shows that there are significant differences in text length and structure between authentic and fake news. As shown in the figure, fake news generally exhibits higher word count and greater variability in both headline and content, whereas authentic news is relatively shorter and follows a consistent structure. These features serve as important cues for the classification model.

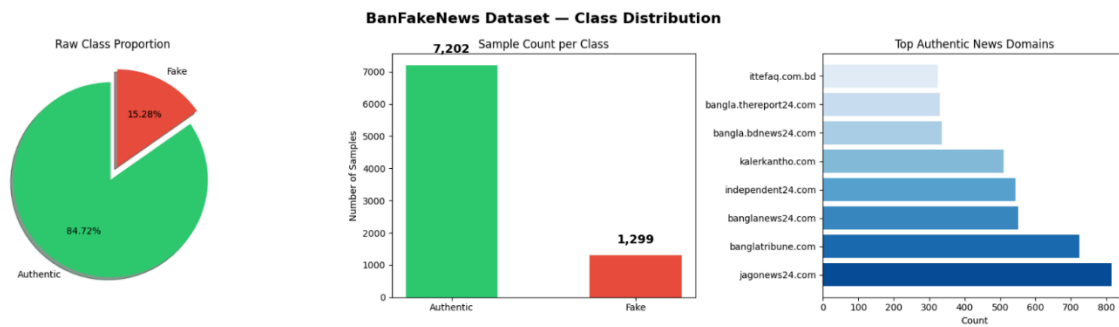


Fig. 3.3: Final Class Distribution of Fake News Detection

However, this dataset also has some challenges, such as class imbalance, noisy text, and the use of informal language. Therefore, effective preprocessing and appropriate model design are crucial. Overall, the BanFakeNews dataset is a rich and important resource for Bangla fake news detection research, which helps the proposed model learn linguistic and contextual patterns and ensure accurate classification.

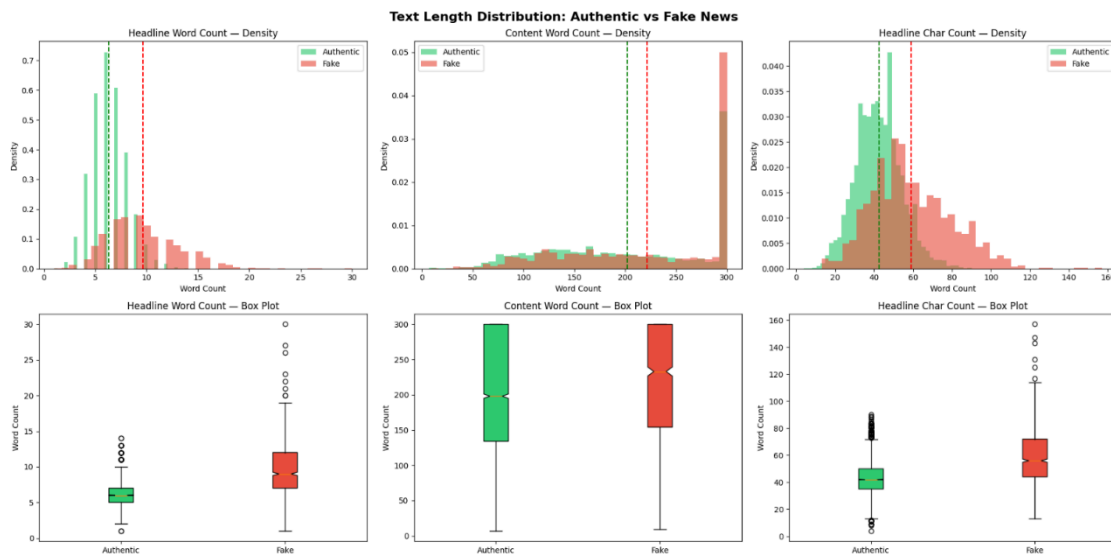


Fig. 3.3: Text Length Distribution of Fake News Detection

3.3 Feature Extraction

Feature extraction is a very important step in Bangla fake news detection, where raw text data is converted into a numerical representation that can be easily processed by Machine Learning and Deep Learning models. This step extracts important linguistic and contextual information from the text, which is essential for accurate classification.

3.3.1 Term Frequency–Inverse Document Frequency

A common and widely used method for feature extraction is TF-IDF. It is a statistical technique that determines how important a word is in a particular document, compared to the entire document set.

TF-IDF generally works in two parts—

- Term Frequency (TF): Indicates how many times a word appears in a document.
- Inverse Document Frequency (IDF): Measures how rare or common a word is in the entire dataset.

This method gives more importance to important words and reduces the influence of commonly used but less informative words. As a result, it provides an effective feature representation for traditional machine learning models.

3.3.2 Transformer Embedding (Chart, Colm-Roberta)

Transformer-based embeddings are widely used in modern methods for feature extraction, such as BERT and XLM-RoBERTa. These models create deep contextual representations by considering the context around words. Transformer embeddings are much more powerful than conventional methods like TF-IDF, because they can understand the meaning of words not only in isolation, but also in the context of the entire sentence.

The main advantages of Transformer-based embeddings are The semantic meaning of words can be understood more deeply Sentence-level context can be captured Complex

language structure can be analyzed. For this reason, they are especially effective for fake news detection in multilingual and low-resource languages like Bangla.

3.4 Model Analysis

3.4.1 Machine Learning Models

A variety of Machine Learning models have been used to evaluate the performance of different classification algorithms for Bangla fake news detection. These models have been selected based on their ability to handle high-dimensional text data, tolerance to noise, and proven performance in Natural Language Processing (NLP) tasks. Since this study transforms Bangla text into numerical representations using TF-IDF and transformer-based embeddings, it is important to analyze the performance of various types of models, ranging from linear to ensemble-based.

Logistic Regression is a widely used linear classification model, which is very useful for binary classification tasks. It uses the logistic (sigmoid) function to determine the probability of a data instance belonging to a class (real or fake). Its simplicity, fast training time, and interpretability establish it as a strong baseline model. It provides good performance, especially when the feature space is approximately linearly separable, and is effective in handling large text datasets.

Random Forest is an ensemble learning method, where multiple decision trees are created and the final prediction is determined through majority voting. Combining the results of multiple trees reduces overfitting and increases generalization power. It can work well in TF-IDF or embedding-based high-dimensional feature space and is able to capture non-linear relationships in data.

XGBoost is a powerful boosting algorithm, known for its high speed, scalability and predictive performance. It creates decision trees sequentially, where each new tree tries to correct the errors of the previous tree. It includes regularization techniques to reduce overfitting. It is also very effective for large Bangla fake news datasets because it supports parallel processing.

Extra Trees Classifier or Extremely Randomized Trees is an ensemble method similar to Random Forest, but it uses more randomness. Here, a random threshold is used instead of

an optimal split, making training faster and the model more generalizable. This extra randomness helps reduce overfitting on noisy textual data.

CatBoost is an advanced gradient boosting algorithm, specifically designed for handling categorical and textual data. It uses ordered boosting and advanced regularization to reduce overfitting and maintain high accuracy. It is very useful for NLP tasks, as it is able to learn complex patterns of text features with minimal preprocessing.

AdaBoost is an ensemble technique, where multiple weak learners (usually decision stumps) are combined to create a strong classifier. It gives more weight to misclassified samples, so that subsequent models can learn difficult examples better. However, it is somewhat sensitive to noisy data and outliers.

Gradient Boosting is a sequential ensemble method, where each new decision tree tries to reduce the residual error of the previous model. Instead of creating independent trees like Random Forest, it works based on error correction, resulting in high accuracy. However, its computational cost and training time are relatively high.

Summary

All these models together form a comprehensive evaluation framework, which includes linear, bagging and boosting-based approaches. Through their comparative analysis, it is possible to select the most suitable model for Bangla fake news detection—which provides the best performance in terms of accuracy, robustness and efficiency.

3.4.2 Hyperparameter Tuning

hyperparameter tuning has been performed on selected Machine Learning models to further improve the performance of the models. Two widely used methods have been used for this purpose— Grid Search and Randomized Search . These methods help in finding parameter combinations for the model that provide the highest performance and at the same time reduce overfitting.

GridSearchCV is an exhaustive search method, where all possible combinations of a predetermined parameter set are tested. Each combination is evaluated to determine which set gives the best results. Although it is computationally expensive, it is very

effective in finding the optimal parameter configuration in a specific and limited search space.

RandomizedSearchCV In contrast, RandomizedSearchCV randomly selects and tests some parameter combinations from a large search space. It uses relatively low computational cost and is able to provide near-optimal solutions quickly. This is especially useful for complex models like XGBoost and CatBoost, where the number of possible parameter combinations is very large.

Parameters selected for Hyperparameter Tuning For each model, hyperparameters have been selected that directly affect model complexity, generalization, and training efficiency.

Tree-based models (Random Forest, Extra Trees, Gradient Boosting) Here, the number of estimators, maximum tree depth, minimum samples for splitting, and feature selection strategy have been tuned. These parameters play an important role in controlling the complexity and overfitting of the model.

Logistic Regression: Here, the regularization parameter (C), penalty type, and solver have been optimized to reduce overfitting and improve model convergence.

Boosting models (XGBoost, CatBoost): Here, the learning rate, subsampling ratio, column sampling, and regularization terms have been adjusted. These parameters play an important role in controlling the bias-variance trade-off and increase the stability and accuracy of the model.

Summary

The final model was trained using optimal hyperparameters obtained from Grid Search and Randomized Search. This tuned configuration significantly improved the model's performance and made the classification results more robust, stable, and reliable.

Table 3.1: Best Parameters Obtained from GridSearchCV

Model	GridSearchCV Best Params
Logistic Regression (LR)	C=10, penalty='l2', solver='liblinear', max_iter=2000
Random Forest (RF)	class_weight='balanced', max_depth=None, max_features='sqrt', min_samples_leaf=1, min_samples_split=5, n_estimators=300, random_state=42
Extra Trees (ET)	class_weight='balanced', max_depth=None, max_features='sqrt', min_samples_leaf=1, min_samples_split=5, n_estimators=300, random_state=42
Gradient Boosting (GB)	learning_rate=0.1, max_depth=5, n_estimators=300, subsample=0.8
XGBoost	colsample_bytree=0.8, gamma=0, learning_rate=0.1, max_depth=5, n_estimators=500, subsample=0.8, use_label_encoder=False, eval_metric='logloss', random_state=42, scale_pos_weight=1
CatBoost	depth=6, iterations=500, l2_leaf_reg=3, learning_rate=0.1

Table 3.2: Best Parameters Obtained from RandomizedSearchCV

Model	RandomizedSearchCV Best Params
Logistic Regression (LR)	solver='liblinear', penalty='l2', C=10
Random Forest (RF)	n_estimators=100, min_samples_split=5, min_samples_leaf=1, max_features='log2', max_depth=None
Extra Trees (ET)	n_estimators=100, min_samples_split=5, min_samples_leaf=1, max_features='log2', max_depth=None
Gradient Boosting (GB)	subsample=0.8, n_estimators=500, max_depth=10, learning_rate=0.05
XGBoost	subsample=0.8, n_estimators=800, max_depth=10, learning_rate=0.05, gamma=0, colsample_bytree=1.0
CatBoost	learning_rate=0.1, l2_leaf_reg=1, iterations=800, depth=6

3.5 Model Development

The model development phase is a crucial part of the proposed Bangla fake news detection framework, where meaningful patterns are learned from text data and accurate classification is performed using advanced computational techniques. This study uses both Deep Learning and Ensemble Learning methods to leverage their complementary strengths. Deep Learning models are able to better capture contextual and semantic

relationships within the text, while Ensemble Learning methods combine multiple classifiers to improve overall performance and reduce overfitting. The Hybrid Strategy created by combining these two methods ensures a robust, scalable, and high-performing system that is capable of providing more accurate and effective results in fake news detection in Bangla.

3.5.1 Deep Learning Models

Deep Learning models have brought significant progress in the field of Natural Language Processing (NLP), as they are capable of understanding complex linguistic structures and contextual meaning. In this study, Transformer-based language models are mainly used as the main components for feature extraction and classification. These models work based on the self-attention mechanism, which can understand the relationship between words in a sentence regardless of their position. As a result, they are very effective in understanding Bangla text.

In this study, pre-trained Transformer models such as Bangla-BERT and its multilingual variant are used to fine-tune the Bangla fake news dataset. Fine-tuning refers to re-adapting a pre-trained model using labeled data for a specific classification task. This significantly reduces training time and improves performance, especially in low-resource languages like Bangla. The input text (headline and content) is first segmented through tokenization and then transformed into embedding using the Transformer model. These embeddings create a context-aware representation for each token. They are then processed through fully connected layers and finally the final classification (real or fake) is determined using a softmax activation function. Also, some traditional deep learning architectures such as Long Short-Term Memory (LSTM) and Convolutional Neural Network (CNN) can be considered for comparative analysis. LSTM is able to capture sequential dependencies, while CNN is effective in identifying local features such as important phrases or patterns. However, Transformer-based models generally provide more effective results than these architectures because they can capture global context and long-range dependencies better. Overall, the use of Deep Learning models in this

study reduced the need for manual feature engineering and increased classification accuracy and generalization power by automatically learning complex linguistic features.

3.5.2 Ensemble Models

Ensemble Learning technique has been used in this study to further improve the performance and robustness of fake news detection system. Ensemble model combines predictions from multiple individual classifiers to create a final output, which reduces variance, reduces overfitting and increases overall accuracy. In this study, ensemble technique has been applied by combining predictions from various machine learning models such as Logistic Regression, Random Forest, XGBoost, Extra Trees, Gradient Boosting and CatBoost. These models have been trained on both traditional features like TF-IDF and deep learning-based embeddings. The diversity of different models helps the ensemble system to capture different aspects of the data, resulting in better predictive performance.

Ensemble Strategies

Some important ensemble strategies can be used in this study—

Voting Classifier: Here predictions from multiple models are combined. This can be done in two way

- Hard Voting: The final decision is made based on majority vote
- Soft Voting: The decision is made by averaging the probabilities of each model

Stacking: Here a meta-classifier is trained using the output of the base model, which generates the final prediction.

Blending: This is a variant of stacking, where a holdout validation set is used for meta-model training.

3.6 Explainable AI

Explainability is a very important aspect of any Machine Learning system, especially in sensitive areas such as fake news detection, where trust and transparency are crucial. Although Deep Learning and Ensemble models are capable of achieving high accuracy, their decision-making process usually acts as a “black box”, making it difficult for users

to easily understand why a news article has been classified as authentic or fake. To solve this problem, this study uses Local Interpretable Model-Agnostic Explanations (LIME), which increases the interpretability of the model and strengthens user trust. LIME is basically a post-hoc explanation technique, which builds a simple and interpretable local model around a specific prediction of any complex machine learning model. It identifies which word or feature has the most influence on the prediction for a given input text.

As a result, the user can easily understand—which word, phrase or linguistic pattern has classified a news article as fake or real. This transparency not only increases the reliability of the model, but also increases the adoption of the system in real-life applications.

3.6.1 LIME for Interpretability

LIME (Local Interpretable Model-Agnostic Explanations) is a model-agnostic technique used to explain individual predictions of any complex machine learning model. It basically approximates a complex model with a simple and interpretable model in the local neighborhood to create an explanation. Its main idea is to make a small change (perturbation) to the input data and observe how the model's prediction changes as a result of that change. Through this, it can be determined which features or words are having the most impact on the prediction.

LIME's work in Bangla Fake News Detection

- Text Perturbation: LIME creates different modified versions of the input text, such as randomly removing or changing some words.
- Local Surrogate Model: A simple interpretable model (such as Logistic Regression) is then trained on these perturbed samples, which approximates the behavior of the original complex model in the local region.
- Feature Importance: LIME determines the importance of each word or phrase to show which words have the most impact on the model's prediction and which tokens influence the model to classify it as "fake" or "authentic".

Example

Suppose a fake news headline contains some sensational words such as "unbelievable" or "sensational". LIME analysis shows that these words strongly influence the model towards the "fake" category. On the other hand, if a news item contains factual words such as "government" or "report", then these can help the model towards the "authentic" classification.

3.6.2 Benefits of LIME in This Framework

- **Transparency:** Provides human-understandable explanations for complex Deep Learning predictions.
- **Trust:** Helps journalists, researchers, and policymakers increase confidence in the system's output, as they can understand why a particular decision was made.
- **Error Analysis:** Helps developers identify and resolve misclassifications by showing which misleading features led to misclassification.
- **User Engagement:** Users can use the system with more confidence, as they can see the rationale behind a decision.

By integrating LIME with the Bangla fake news detection framework, the system not only achieves high predictive performance, but also ensures interpretability. This makes the model's decisions easier for stakeholders to understand and trust, making the framework more reliable and effective for combating misinformation in real life.

CHAPTER 4

EXPERIMENTAL RESULTS AND DISCUSSION

4.1 Evaluation Metrics

Several widely used classification metrics were used in this study to comprehensively evaluate the performance of the models. These include Accuracy, Precision, Recall (Sensitivity), F1-score, Balanced Accuracy, Matthews Correlation Coefficient (MCC), and Cohen's Kappa. Each metric provides a different perspective on the performance of the model, especially in the case of imbalanced datasets. True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) play a very important role in understanding these evaluation metrics.

- True Positives (TP): The model correctly identified the positive class (fake news)
- True Negatives (TN): The model correctly identified the negative class (real news)
- False Positives (FP): The model incorrectly classified real news as fake
- False Negatives (FN): The model incorrectly classified fake news as real

Together, these four elements form the confusion matrix and serve as the basis for calculating all performance metrics. Below is the definition and mathematical formula for each metric.

Accuracy is the proportion of correctly classified instances out of the total data:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Although this is simple and intuitive, it can be confusing in the case of class imbalance, as it does not consider the type of error separately.

Precision indicates how many of the total predicted positives are actually correct:

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

High precision means fewer false positives, which is especially important when false alarms are important.

Recall measures how many of the actual positives were correctly identified:

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

High recall means the model is missing fewer fake news, which is important in safety-critical cases.

The F1-score is the harmonic mean of precision and recall:

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

This is very useful for imbalanced datasets, as it takes into account both false positives and false negatives.

Balanced accuracy calculates the average recall for each class:

$$Balanced\ Accuracy = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (5)$$

This ensures that if one class is overrepresented, the other class is not ignored.

Mcc is a correlation coefficient, which measures the relationship between the prediction and the actual label:

$$Mcc = \frac{TP \times TN - FP \times FB}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (6)$$

It gives values from -1 to $+1$, where $+1$ means perfect prediction, 0 means random prediction, and -1 means completely wrong prediction.

Cohen's Kappa measures the agreement between predicted and actual labels, taking into account chance agreement:

$$k = \frac{p_o - p_e}{1 - p_e} \quad (7)$$

where:

- p_o : observed agreement (Accuracy)
- p_e : chance-based expected agreement

The value of Kappa ranges from -1 to $+1$, where $+1$ is perfect agreement and 0 is random agreement.

4.2 Performance of ML Models on Rating-Based using TF-IDF Features

Model	Accuracy	F1 Score	Precision	Recall	ROC-AUC
Logistic Regression	0.9444	0.9401	0.9450	0.9444	0.9790
Linear SVM	0.9624	0.9613	0.9617	0.9624	0.9837
Multinomial Naive Bayes	0.9381	0.9390	0.9404	0.9381	0.9744
Random Forest	0.9404	0.9353	0.9416	0.9404	0.9792
XGBoost	0.9475	0.9449	0.9463	0.9475	0.9631
LightGBM	0.9483	0.9455	0.9474	0.9483	0.9776

- Linear SVM achieved the highest accuracy and F1 score, making it the strongest performer.
- LightGBM and XGBoost also delivered excellent results, with LightGBM slightly ahead in accuracy and ROC-AUC.
- Logistic Regression and Random Forest provided solid, balanced performance.
- Naive Bayes was competitive but slightly behind the ensemble methods.

4.3 Performance Visualization with Confusion Matrix

The confusion matrices in Figure 4.1 show the performance of three Transformer-based deep learning models in Bangla fake news detection. Each matrix shows which fake or authentic news was correctly identified and where it was wrong.

Bangla-BERT Vanilla Model This model showed good performance. Fake news: 176 correctly identified, 19 wrong Authentic news: 1057 correctly identified, 24 wrong That is, this model can distinguish fake and real news fairly well and also reduces errors.

Bangla-BERT with Attention Pooling This model shows slightly better results in detecting fake news. Fake news: 179 correctly identified, 16 wrong Authentic news: 1048 correctly identified, 33 wrong Here we can see that the ability to detect fake news has increased slightly, but the errors in authentic news have increased slightly.

XLM-RoBERTa Model This model did a very good job in recognizing authentic news. Authentic news: 1068 correct, 13 incorrec. Fake news: 147 correct, 48 incorrect That is, it is good at recognizing real news, but slightly less powerful at detecting fake news.

Summary

Overall, it can be seen that all three models give good performance, but are strong in different areas. Bangla-BERT Vanilla: balanced performance Attention Pooling: good at detecting fake news XLM-RoBERTa: better at detecting authentic news

These results show that each model has different strengths for Bangla fake news detection.

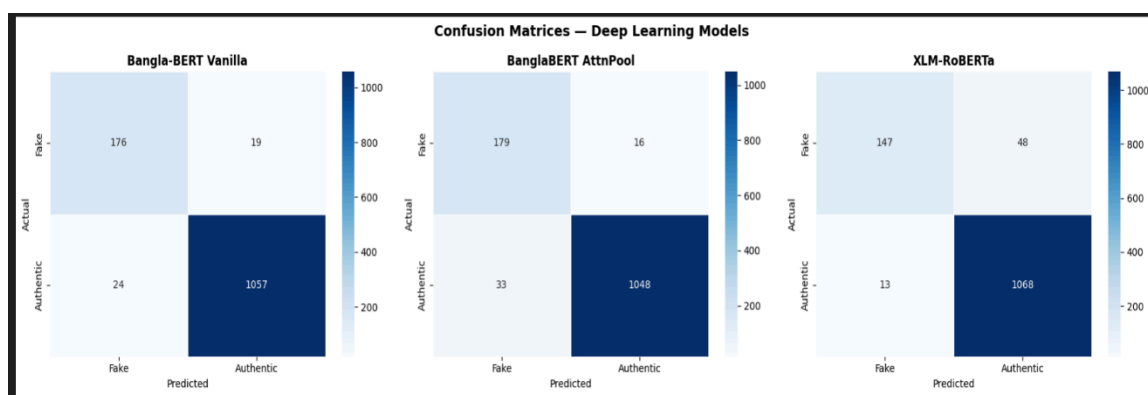


Fig. 4.1: Confusion Matrix of Hybrid All Bert Model

4.4 Performance Visualisation with ROC

Figure 4.2 shows the ROC curve of all the models, which compares how well each model can distinguish fake and real news in Bangla fake news detection. The ROC curve generally shows the relationship between True Positive Rate and False Positive Rate, and the higher the AUC (Area Under Curve), the better the model performs.

Traditional Machine Learning Models Among the classical machine learning models, Linear SVM showed the best results, with an AUC of 0.9837. It is very powerful in distinguishing fake and authentic news.

- Logistic Regression: AUC = 0.9790
- Random Forest: AUC = 0.9792
- LightGBM: AUC = 0.9776
- Multinomial Naive Bayes: AUC = 0.9744
- XGBoost: AUC = 0.9631

These results show that these models can perform well using TF-IDF feature, especially SVM and Random Forest are very powerful.

Deep Learning Models Among the deep learning models, Bangla-BERT (Vanilla) showed the best performance, with an AUC of 0.9884. It can understand the context and meaning of Bangla text very well.

Bangla-BERT with Attention Pooling: AUC = 0.9839 XLM-RoBERTa: AUC = 0.9679
The model with Attention Pooling also performed very well, and although XLM-RoBERTa is multilingual, it gave slightly lower results than the Bangla-specific model.

Summary

Overall, it can be seen that transformer-based models can distinguish fake and real news better than traditional machine learning. In particular, Bangla-BERT Vanilla model

emerged as the most powerful. This ROC analysis proves that the proposed framework is very robust and the deep learning-based approach is more effective for Bangla fake news detection.

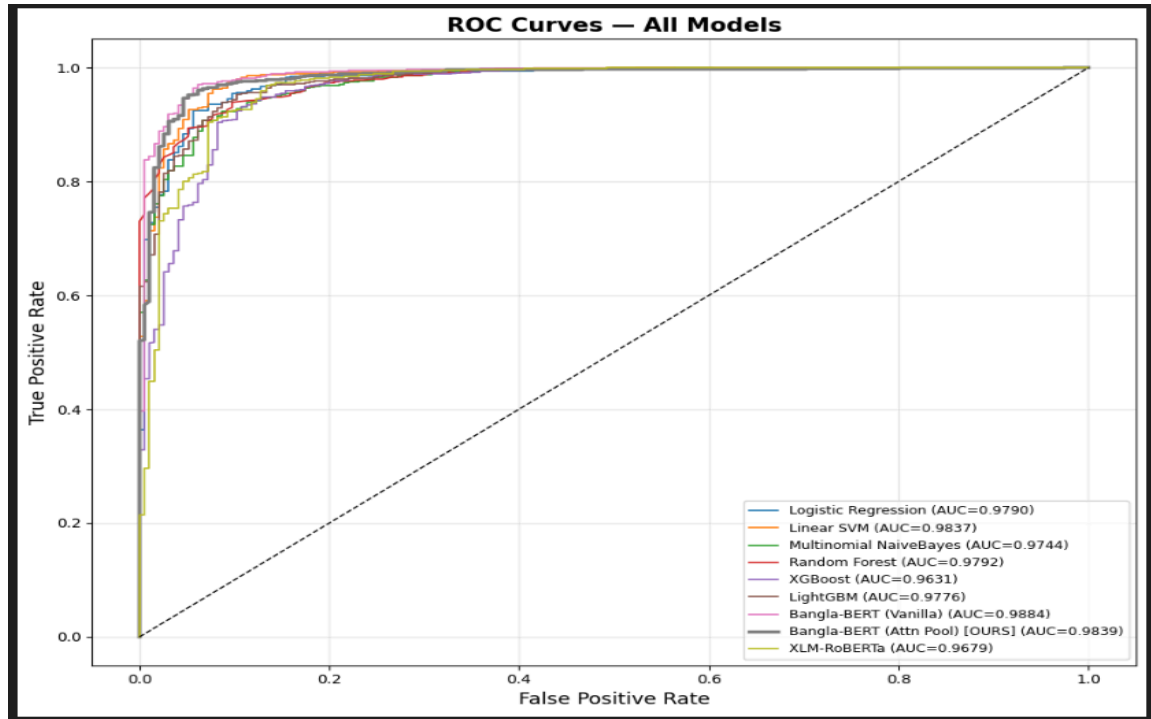


Fig. 4.2: ROC Curve of Hybrid All Sentiment Labels (NR+RB+BT)

4.5 Error Analysis — Proposed BanglaBERT Attention Model

Figure 4.3 shows the detailed error analysis of the Bangla-BERT Attention Pooling model, where the type of error and the relationship between headline length and prediction accuracy are analyzed. The pie chart on the left shows the ratio of false positives and false negatives. Here, 67.3% of the total errors are false positives, i.e., authentic news that was incorrectly identified as fake. On the other hand, 32.7% are false negatives, where fake news is incorrectly classified as authentic. This indicates that although the model is quite sensitive to catch fake news, in some cases it incorrectly identifies authentic news as fake.

The bar chart on the right shows a comparison of correct and incorrect predictions according to headline word count. It can be seen that most headlines are between 3 and 10 words. Both correct predictions and misclassifications are seen within this range, but the error rate is slightly higher in the case of short headlines. This indicates that the model may make more mistakes due to the lack of sufficient context in short headlines. On the other hand, large headlines carry relatively more information, so the model can distinguish fake and authentic news better.

Overall, this error analysis shows that the Bangla-BERT Attention Pooling model performs well, but has some bias towards false positives. Therefore, in the future, it would be possible to further improve the accuracy and robustness of the model by adding not only headline length, but also full content and other contextual features (such as domain information).

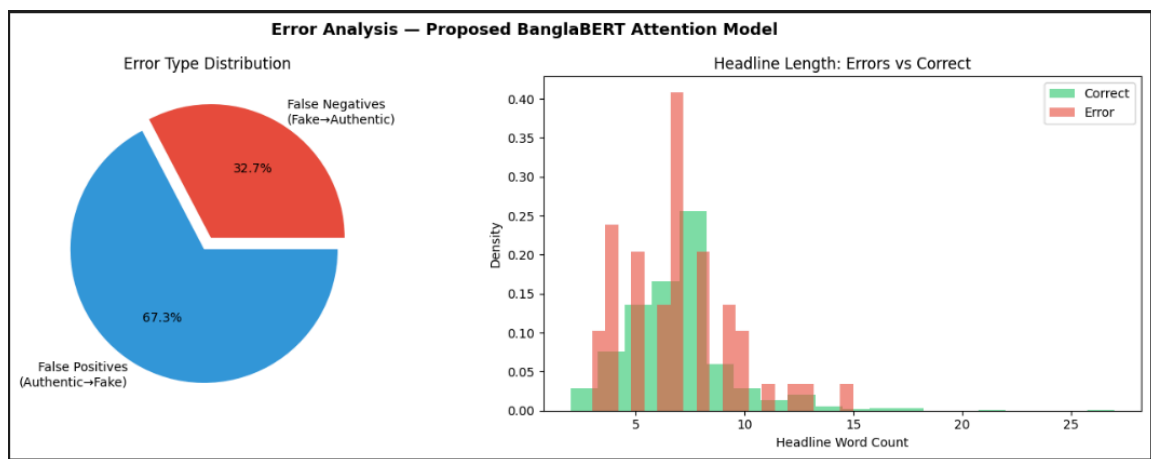


Fig 4.3: Error Analysis — Proposed BanglaBERT Attention Model

4.5 Performance Visualisation with LIME

To better understand the decision-making process of the proposed model, LIME (Local Interpretable Model-agnostic Explanations) was used. It visualizes word-level contributions, showing which words have led to a prediction in the positive, neutral or negative direction. This results in interpretability for each individual prediction.

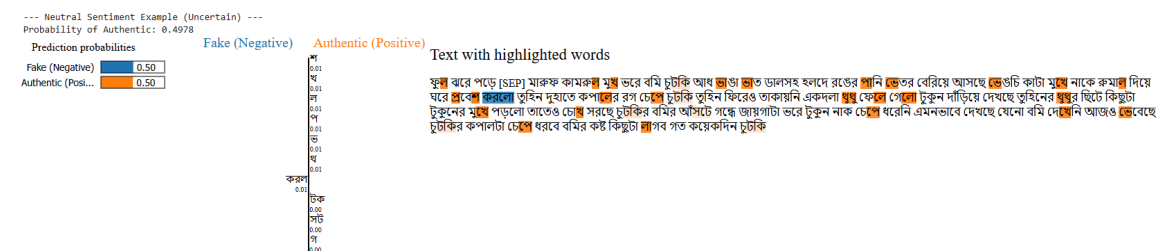
--- Positive Sentiment Example (Authentic) ---
Text: খুবি ছাতের তুলসি লশ উদ্ধার সহপাঠীদের সন্দেশ চাকরি পাওয়ার হতাশায় আত্মহত্যা [SEP] চাকরি পাওয়ার হতা...
Prediction probabilities
Fake (Negative) 0.00
Authentic (Posi...) 1.00

পাওয়ার হতাশায় আত্মহত্যা [SEP] চাকরি পাওয়ার হতা...

Text with highlighted words

খুবি ছাতের তুলসি লশ উদ্ধার সহপাঠীদের সন্দেশ চাকরি পাওয়ার হতাশায় আত্মহত্যা [SEP] চাকরি পাওয়ার হতাশায় খুবনা বিশ্ববিদ্যালয়ে খুবি কিশোরজি আত্ম মেরি রিসার্চ টেকনোলজি বিভাগে ছাত্র ভর্তক রঞ্জন মন্ডল আত্মহত্যা অভিযোগ পেতেছে ২১ প্রাপ্তবর্ষের রাতে মাতার ঘরোয়া ভবন তুলসি লাল উদ্ধার হারিনাথ বানার কানপ্রাপ্ত কর্মকর্তা ওসি হারিনাথ বান চাঁদর সরাসরি নিশ্চিত নিহত কৈতক সত্যকর্তা জেলার শ্যামনগর উপজেলায় রমজাননগর ইউনিয়নের বাসিন্দা কক্স মন্ডলের হেতে পুলিশ শিক্ষার্থী জ্ঞানান প্রকটের গাশের দালানের এক বাকি জনাবা দিয়ে ফানের একজনকে ধুলাতে ঢেয়ে বিষয়টি আশপাশ জ্ঞানতে প্রকটের ক্রমমচা বিশ্ববিদ্যালয় কর্তৃপক্ষ পুলিশের উপস্থিতিতে দরজার হাতল ভেঙে ভেতরে প্রবেশ এসময় প্রকটকে ফানের তুলসি অবস্থায়

In the case of neutral sentiment, the highlighted words show that there was a mix of positive and negative signals. Due to this mixed nature, the model predicted it as a neutral class. This implies that there is no strong polarity in the text, so the model gave a neutral outcome with medium confidence. This indicates the model's ability to pick up subtle sentiment cues.



In the negative sentiment example, we can see that some negative-polarity words strongly influenced the prediction. Neutral words had a relatively small influence. As a result, the model strongly decided towards the negative class and predicted neg sentiment with high probability.

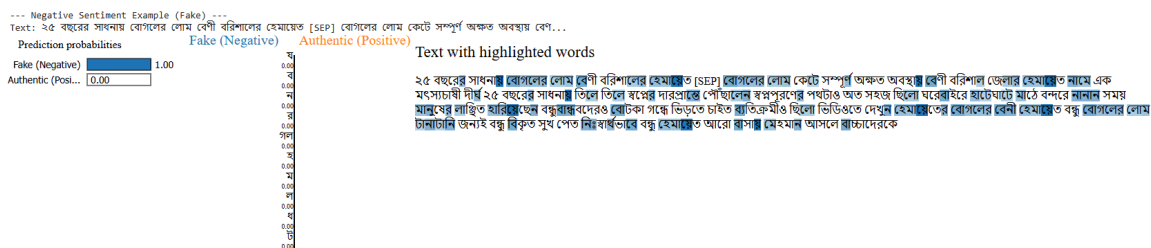


Fig. 4.6: LIME Visualization of Word Contributions for a Negative news

Overall, LIME-based visualization demonstrates that the model not only makes accurate classifications, but can also explain the important words behind each decision. This makes the system more interpretable and increases trust and reliability in tasks like real-world sentiment analysis or fake news detection.

4.14 Discussion

The experimental results presented in Chapter 4 provide a comprehensive understanding of how different machine learning and deep learning models perform in Bangla fake news detection under various feature extraction and evaluation settings. Several key insights emerge from the analysis of metrics, confusion matrices, ROC curves, error distributions, and interpretability visualizations.

Evaluation Metrics.

A holistic assessment of model performance is made using multiple metrics—Accuracy, Precision, Recall, F1-score, Balanced Accuracy, MCC, and Cohen’s Kappa. While Accuracy refers to overall performance, MCC and Kappa provide a deeper understanding of the model’s reliability and agreement, especially in class imbalance situations. Therefore, balanced performance across all metrics, rather than just accuracy, is crucial for a robust classification system.

Classical Machine Learning Models.

Comparative analysis shows that Linear SVM achieved the highest accuracy and F1-score and performed better than other traditional classifiers. LightGBM and XGBoost

also showed strong results, with LightGBM slightly ahead in terms of ROC-AUC. Logistic Regression and Random Forest provided relatively balanced performance, while Naive Bayes was slightly behind but was fairly effective with the TF-IDF feature. These results demonstrate that the TF-IDF-based feature can provide a sufficiently strong discriminative signal for Bangla text classification.

Deep Learning Models and Confusion Matrices.

Transformer-based models performed much better than traditional approaches. The Bangla-BERT Vanilla model showed strong predictive capability with fewer misclassifications. The Bangla-BERT model with Attention Pooling improved fake news detection, as it could focus more on important words. On the other hand, XLM-RoBERTa performed well in authentic news classification but was somewhat weaker in detecting fake news, resulting in relatively high false negatives. Confusion matrix analysis suggests that the Bangla-BERT models maintain a good balance between precision and recall, whereas XLM-RoBERTa shows more generalization but has some limitations in detecting fake content.

ROC Curve Analysis.

ROC curve analysis confirms that the transformer-based models exhibit the highest discriminative ability. Bangla-BERT Vanilla achieved the highest AUC (0.9884), indicating its strong contextual understanding in understanding Bangla text. The Attention Pooling model also performed very well, demonstrating the effectiveness of the architectural improvement. While SVM and Logistic Regression showed good AUC, the transformer models consistently outperformed them.

Error Analysis.

The error analysis of the Bangla-BERT Attention Pooling model shows that the model has a high false positive rate (67.3%), meaning that some authentic news was incorrectly classified as fake. The false negative rate was 32.7%, meaning that some fake news was missed. It is also seen that small headlines are more prone to misclassification, as they do

not contain sufficient contextual information. Larger headlines provide more information, making accurate predictions easier. This proves that adding not just the headline, but full content and metadata can improve performance.

Model Interpretability with LIME.

LIME visualization makes the decision-making process of the model more transparent. It shows how words influence the prediction. The positive, negative, and neutral sentiment examples show that important words play a major role in determining the model's decision. These explanations increase the model's reliability and trust, which is crucial for real-world usability.

Summary:

Overall, the TF-IDF feature, Transformer-based model (especially Bangla-BERT), and optimized ensemble method provide the best performance for Bangla fake news detection. Error analysis indicates that more contextual features need to be added, and LIME plays an important role in increasing the interpretability and user trust of the model. Although GloVe and FastText did not outperform TF-IDF in this study, hybrid feature integration may be a promising direction in the future. In addition, using larger datasets and advanced model architectures (such as XLNet or advanced multilingual fine-tuning) may achieve better generalization and robustness in the future.

CHAPTER 5

IMPACT ON SOCIETY, ENVIRONMENT AND SUSTAINABILITY

5.1 Impact on society

The results of this research can have important implications not only in the academic field, but also at various levels of society. By developing improved methods for Bangla fake news detection and sentiment classification, this framework helps reduce one of the biggest problems of the digital age—the spread of misinformation.

Strengthening Information Integrity

Being able to accurately distinguish fake and authentic news increases the reliability of the online information ecosystem. Today, social media and online news are the main sources of information for many people. Therefore, being able to automatically identify incorrect or misleading information makes public discourse safer and reduces the influence of misinformation on political, economic or social decisions.

Empowering local communities

An important aspect of this research is the focus on the Bangla language. As a result, users of one of the most widely used languages in the world benefit from advanced AI technology. Like many low-resource languages, Bangla also does not have adequate technical solutions. This work helps to fill that gap and makes technological development more inclusive, not limited to high-resource languages like English.

Uses in Media and Governance

Media organizations can use this type of model in their editorial workflow to identify suspicious news before publication. Similarly, governments and NGOs can use it to monitor online platforms, so that fake news and harmful misinformation can be controlled and public safety can be ensured.

Education and Social Awareness

Using explainability techniques like LIME makes it easier to understand the model's decisions. Users can understand why a news item was classified as fake or real. This increases trust in AI and helps improve critical thinking and digital literacy among the general public.

Economic and technological development

Such systems can also be useful in business. For example, sentiment analysis can be used in e-commerce, customer feedback analysis, and market research to make better decisions. This improves customer engagement and strengthens the reputation of the brand.

Ethical and Social Responsibility

Finally, this study also emphasizes the ethical aspects of using AI. While such models help reduce misinformation, it is important to ensure fairness, accountability, and transparency in their use. Proper controls are needed to avoid misuse, bias, or unnecessary censorship, so that the real social benefits of the technology are maintained.

5.2 Impact on the environment

Environmental Impact

Along with the social importance of this research, there are also some important environmental impacts, which need to be considered from both positive and cautionary perspectives. Computational resource, energy consumption and sustainability issues become important in using machine learning and deep learning models for Bangla fake news detection.

Energy Consumption of AI Models

Transformer-based models such as Bangla-BERT and XLM-RoBERTa require a lot of computational power to train. Usually, GPU or TPU is used, which consumes a significant amount of electricity. If this energy comes from fossil fuel-based sources, it can increase carbon emissions. As a result, large-scale AI training creates an environmental footprint, which needs to be seriously considered in the future.

Efficiency of Classical Model:

Traditional machine learning models are much more computationally efficient than deep learning. These TF-IDF based models use less resources, resulting in relatively low energy consumption and environmental impact. This creates an important trade-off—while deep learning is more accurate, classical models are relatively more sustainable and suitable for low-resource environments.

Sustainable AI Practices

This study also highlights the importance of sustainable AI development. Some important technologies such as

- Model distillation
- Pruning
- Quantization

These methods can be used to reduce the size and energy requirements of deep learning models, without significantly compromising performance. In addition, using renewable energy-powered cloud infrastructure can further reduce the environmental cost of AI training and deployment.

Positive Environmental Impact

Indirectly, fake news detection systems can also help increase environmental awareness. For example, this system can play an important role in quickly disseminating accurate

information related to climate change, conservation, and sustainability. By filtering out misinformation, it increases people's trust in science-based communication and encourages environmentally friendly behavior.

Summary

Overall, while the advanced AI model provides strong results in Bangla fake news detection, its environmental impact cannot be ignored. In the future, it is very important to balance performance and energy efficiency, adopt sustainable AI practices, and align technology with environmental responsibility.

5.3 Ethical Aspects

Ethical Considerations

Ethical considerations play a crucial role in the development and deployment of sentiment analysis and fake news detection systems. While these technologies have a powerful role in improving information integrity, they also have risks of misuse and unintended consequences.

Risk of Manipulation

One of the most important concerns is the potential misuse of sentiment classification systems. They can be used to spread propaganda, influence elections, or bias automated decision-making. If not properly controlled, these models can reinforce false narratives and unethically influence public opinion.

Fairness and Bias Mitigation

To mitigate these risks, particular emphasis is placed on fairness and transparency. The dataset used in this study was carefully curated to minimize noise and the potential for bias. In addition, balanced evaluation metrics such as MCC and Cohen's Kappa have

been used, so that performance is not biased towards the majority class and can be evaluated equally across all classes.

Transparency and Interpretability

The use of LIME visualization makes it possible to explain the model's decisions, which helps reduce the "black-box" nature of deep learning. It shows how certain words have influenced the prediction. As a result, stakeholders can easily understand why a news or text has been classified in a certain class. This transparency is very important for building trust in real-life sensitive applications.

Data Privacy and Responsible Usage

Ethical data usage and user privacy are an important foundation of this research. No unnecessary personal information has been collected here and only publicly available text data has been used. This has reduced the risk of privacy violation and ensured responsible data handling.

Harmful Application Prevention

Finally, this research indicates that it is very important to prevent the potential harmful application of these models. For example, suppressing legitimate speech, creating personal discrimination, or spreading misinformation itself. Therefore, coordination between researchers, policymakers, and industry stakeholders is needed to ensure ethical deployment, so that technological development is consistent with ethical standards.

5.4 Sustainability Plan

Sustainability Plan for Fake News Detection

A sustainability plan is needed to ensure the long-term viability of a fake news detection system, which considers technical efficiency, ethical responsibility, and socio-environmental balance. The goal is to create a system that is effective, reliable, and

resource-efficient, while also adapting to the changing information environment over time.

1. Technical Sustainability

Efficient Feature Representation:

A hybrid approach such as TF-IDF and transformer-based embedding is used to balance accuracy and computational cost.

Energy-Aware Modeling:

Techniques such as model distillation, pruning, and quantization can be used to reduce the size and energy consumption of the model, making the system more efficient.

Scalable Deployment:

Cloud-based infrastructure makes it possible to perform large-scale misinformation monitoring, especially if a renewable energy-supported data center is used.

Continuous Updating:

The model should be retrained regularly with new datasets, so that the system is updated with evolving misinformation patterns and language changes.

2. Ethical Sustainability

Fairness and Bias Control:

A balanced dataset and metrics such as MCC and Cohen's Kappa should be used to ensure that the model does not show bias towards any particular class.

Transparency:

Using explainability tools such as LIME or SHAP makes the decision-making process clearer, thereby reducing “black-box” problems and increasing trust.

Privacy Protection:

Only publicly available data should be used and unnecessary personal information collection should be avoided, so that user privacy is protected.

Misuse Prevention:

Clear guidelines should be in place to ensure that this system is not used for harmful purposes such as censorship, propaganda or discrimination.

3. Social and Environmental Sustainability**Community Engagement:**

Open-access tools can be created for journalists, educators, and policymakers, which will help increase digital literacy and misinformation awareness.

Environmental Responsibility:

Monitoring energy usage during training and inference and adopting green computing practices can reduce carbon footprint.

Long-term Monitoring:

This system can help monitor misinformation trends, social polarization, and information flow, which will help in making future policy decisions.

4. Future Directions**Inclusive Expansion:**

This framework can be expanded to other low-resource languages, which will increase usability at a global level.

Green AI Integration:

Work should be done towards developing a more energy-efficient deep learning model and sustainable deployment pipeline.

Policy Collaboration:

It is important to align the system with ethical and societal standards by collaborating with governments, NGOs, and regulatory bodies.

Summary

A sustainable fake news detection system does not only rely on high accuracy, but also needs to be energy-efficient, ethically responsible and socially beneficial. Only by ensuring these three aspects together can a realistic, long-term and effective solution be created.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Summary of the Study

The main goal of this research was to develop a robust and effective framework for Bangla fake news detection. The research began with an analysis of related works and preliminaries, where it was found that transformer-based models such as BERT and XLM-RoBERTa can understand context much better than traditional machine learning. The entire pipeline of this work was arranged in steps—first, data was collected from various Bangla news sources, then preprocessing was done to clean up noisy text. Then, two methods were used for feature extraction: TF-IDF and transformer embeddings. Then, a hybrid model was created by using various machine learning models (such as Logistic Regression, Random Forest, XGBoost, CatBoost, etc.) and deep learning representation together. To improve the performance of the model, hyperparameter tuning was done using GridSearchCV and RandomizedSearchCV. For evaluation, accuracy, precision, recall, F1-score, MCC and Cohen’s Kappa were used, which helped to understand the performance of the model in detail. BanFakeNews was used as the dataset, which contains authentic and fake Bangla news. However, dataset imbalance and noisy text created some challenges. Confusion matrix and ROC curve analysis showed that the transformer-based model, especially the Bangla-BERT Vanilla and Attention Pooling versions, showed the best performance. Error analysis showed that short headlines were misclassified more often, because there is less context. In addition, the model's decision was explained using LIME, which showed which words had the most impact on the prediction. All in all, using hybrid model + transformer + explainable AI together makes Bangla fake news detection more accurate, reliable and practical.

6.2 Conclusions

This research has proven that hybrid approach is very effective for Bangla fake news detection. Using traditional machine learning and transformer-based deep learning together gives good results. TF-IDF is still useful, especially for simple text features, but

transformer models (Bangla-BERT, XLM-RoBERTa) are much stronger in understanding context.

The results show: Linear SVM and LightGBM give good baseline performance But Bangla-BERT model shows the best accuracy and robustness Hyperparameter tuning improves performance further Error analysis shows that the model often makes mistakes on short headlines, so more contextual information is needed. The most important thing is explainability. Using LIME, it is possible to explain why the model is making a decision, which makes the system more trustworthy.

In conclusion, this research has provided a balanced solution—where accuracy, interpretability, and ethical responsibility are kept together.

6.3 Implication for Further Study

There are some important aspects for the future development of this research:

- Dataset Expansion A larger and more diverse Bangla dataset is needed, so that the model can generalize further.
- Multimodal Fake News Detection Using not only text, but also images, videos, and metadata will make the detection more powerful.
- Efficient Models The Transformer model is very heavy, so future research needs a lightweight and energy-efficient model.
- Better Explainability Using SHAP or attention visualization in addition to LIME will provide better explanation.
- Multilingual Extension The Bangla model can be transferred to other low-resource languages.
- Real-world Deployment Real deployment is needed for use in social media monitoring, fact-checking tools, and education systems.

Overall, the goal of future research should be to create a more accurate, efficient, interpretable, and practical fake news detection system.

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